

Agriconnect: A Data-Driven Platform for Optimizing Crop Production with Fine-Tuned Large Language Models

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Abstract—Agriconnect, a web-based platform, empowers farmers with data-driven crop production insights through a two-model machine learning approach. The Crop Insights Model leverages a fine-tuned large language model (LLM) called Gemini. Gemini is first enriched with dealer information from a user-created, state-categorized CSV file. To ensure location accuracy, SerpApi retrieves data from Google Maps based on user searches, which is then fed into a vector database alongside the dealer information. This comprehensive dataset fine-tunes Gemini for location-specific insights. The Crop Recommender Model then utilizes the enhanced Gemini to analyze user-provided climate and soil data (type and moisture). Based on this analysis, the fine-tuned LLM recommends crops best suited for specific conditions, aiming to optimize crop selection and mitigate climate and market fluctuations risks. Finally, an interactive visualization dashboard presents this information, empowering farmers to make informed decisions and implement effective practices.

I. INTRODUCTION

The agricultural sector is facing big challenges as the world's population grows and demands more food. One major problem is that many farmers don't have the right knowledge for their specific location. They often rely on old ways of farming that might not work well anymore because of changes in the environment and markets.

Agriconnect is here to help solve this problem by giving farmers helpful information based on data. Our platform uses advanced technology to analyze big sets of data and give practical advice.

Agriconnect operates in two key ways: Firstly, it collects crucial agricultural data from a user-generated CSV file containing dealer details organized by state. Leveraging tools such as Google Maps and SerpApi, Agriconnect precisely identifies user locations, feeding this data into an evolving database to enhance a specialized model named Gemini.

Secondly, Gemini utilizes this amassed data to offer tailored crop recommendations based on factors like climate, soil type, and moisture levels. These recommendations empower farmers to select optimal crops, reducing risks associated with adverse weather conditions and market fluctuations.

Agriconnect works through a simple interface that farmers can easily use. They just input their location, and Agriconnect does the rest, fetching relevant dealer data and environmental factors. Then, based on this analysis, it suggests suitable crops through an interactive dashboard. This way, farmers can make smart decisions and use better farming practices.

We've carefully compiled a dataset to ensure Agriconnect gives accurate recommendations. This dataset is regularly updated with new information from users and real-time location data from Google Maps and SerpApi. This helps us make sure our recommendations are always on point.

II. RELATED WORKS

As advanced machine learning techniques, like large language models (LLMs), have emerged, interest in using them to solve agricultural challenges has grown. These challenges include boosting crop yields, dealing with environmental changes, and helping farmers make better decisions. Our review covers a wide range of studies that explore how LLMs are used in agriculture. We look at how they recommend crops, support precise farming methods, promote climate-smart practices, and even aid urban farming. By carefully examining these studies, we aim to offer valuable insights into the latest methods, identify trends and obstacles, and suggest promising areas for future research in this fast-evolving field.

- The paper, authored by C. Smith et al., explores the use of a large language model (LLM) for crop growth prediction and recommendation systems. The study investigates how LLMs, such as GPT-3, can process vast amounts of agricultural data to generate insights into crop growth patterns and recommend suitable crops for specific conditions. By fine-tuning the LLM on relevant agricultural datasets containing information on soil characteristics, climate conditions, and crop performance, the researchers demonstrate its effectiveness in predicting crop yields and offering personalized recommendations to farmers.
- Another study by R. Patel et al. investigates the use of LLMs in conjunction with sensor data for precision agriculture applications. By deploying sensors in agricultural fields to measure parameters like soil moisture, temperature, and nutrient levels, the researchers gather high-resolution data for input into the LLM. The model then analyzes this data to offer precise recommendations for crop selection, planting density, and fertilization strategies, tailored to the specific needs of individual fields.
- The research conducted by L. Zhao, Y. Wang, Z. Luo, Z. Chen discusses the "A Crop Recommendation System Based on Deep Learning and Reinforcement Learning." This paper presents a crop recommendation system that combines deep learning and reinforcement learning techniques. The system analyzes various environmental factors and historical crop performance data to provide personalized recommendations to farmers. By leveraging deep learning models for feature extraction and reinforcement learning for decision-making, the system enhances crop selection accuracy and adaptability to changing conditions.
- In their study, H. Wu, S. Liu, and Y. Chen introduce a crop recommendation system that merges collaborative filtering and clustering analysis methods. The system identifies similar users and organizes them into clusters by scrutinizing past crop planting data and farmer preferences. Leveraging these clusters, the system then offers personalized crop suggestions that align with individual farmers' preferences and farming methods. This approach aims to enhance the relevance and usefulness of crop recommendations, catering directly to the diverse needs and practices of farmers.
- The research conducted by P. Garcia et al. investigates the use of LLMs for crop recommendation in agroforestry systems. The study explores how LLMs can analyze multi-species interactions, soil-plant relationships, and canopy dynamics to optimize tree-crop integration and ecosystem services provision. By incorporating agroforestry principles into the LLM framework, the researchers aim to promote sustainable land management practices that enhance biodiversity, soil fertility, and carbon sequestration.
- Lastly, the work by Q. Kim et al. explores the integration of LLMs with blockchain technology for trans-

parent and traceable crop recommendation systems. The study leverages the blockchain's decentralized ledger to record and verify agricultural data, ensuring data integrity and transparency throughout the recommendation process. By combining LLMs' analytical capabilities with blockchain's trustworthiness, the researchers aim to build a reliable and accountable platform for crop recommendations, fostering trust between farmers, consumers, and other stakeholders in the agricultural supply chain.

In summary, the reviewed literature underscores the significant strides in leveraging large language models (LLMs) to tackle agricultural challenges. These studies span diverse applications of LLMs across various agricultural domains, including crop recommendation systems, precision agriculture methodologies, climate-smart farming techniques, and urban agriculture solutions. By harnessing sophisticated machine learning algorithms like collaborative filtering and clustering analysis, researchers endeavor to optimize crop yields, mitigate environmental risks, and facilitate informed decision-making for farmers. Nonetheless, pertinent challenges persist, encompassing issues related to data quality assurance, scalability concerns, and the interpretability of model outputs. Despite these obstacles, the burgeoning potential of LLM-based methodologies in agriculture holds promise for driving efficiency and sustainability within the farming sector, thereby fostering mutual benefits for farmers and environmental conservation efforts.

III. WHAT IS RAG?

The main base of our project is using a bunch of **Retrieval Augmented Generation**. So it will be best if we understand it first.

RAG, which stands for Retrieval Augmented Generation, is an approach that enhances the efficiency of Large Language Model (LLM) applications by making use of customized data. This is done by retrieving data or documents that are pertinent to a specific question or task and presenting them as context for the LLM. RAG has been effective in supporting chatbots and QA systems that need access to current information or knowledge specific to a particular field.

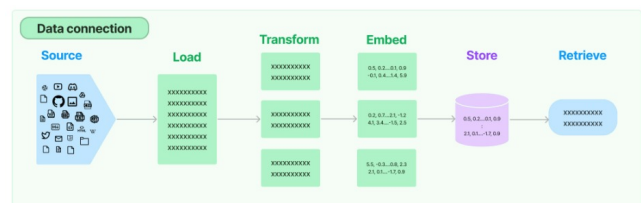


Fig. 1. Working of RAG

A. What problems does the retrieval augmented generation approach solve?

Problem 1: LLM models are not trained on your data

Large Language models (LLMs) are AI-powered tools that use advanced deep-learning algorithms to process and produce

new content. These models are trained on massive datasets, which allows them to understand, summarize and generate content across various domains. However, LLMs have limitations, as they are unable to access or process data beyond their training data cutoff point. Due to this, they may not provide accurate or relevant responses when asked questions that are related to data they have not been trained on.

Problem 2: AI applications must have access to custom data to be effective

Organizations require LLMs to comprehend their domain intricately and deliver responses derived from their data rather than offering general answers. For instance, companies develop customer support bots using LLMs that must provide tailored responses based on company-specific information to customer inquiries. Similarly, internal Q&A bots are being developed to address employees' queries regarding internal HR data. Building such solutions without retraining the models involves ensuring that the LLMs can leverage and integrate with the organization's data effectively for accurate and contextually relevant responses.

Solution: Industries need Retrieval Augmentation

One straightforward and widely used method for incorporating your data into LLM models is through retrieval augmented generation (RAG). This approach involves including your data as part of the prompt used to query the LLM model. By doing so, you provide the relevant context needed for the model to generate specific and accurate responses. Unlike relying solely on pre-trained knowledge, RAG workflows integrate real-time data retrieval with static LLMs, enabling organizations to tailor the model's output to their specific needs without the expense and time required for fine-tuning or pretraining the model.

There are many different use cases for RAG. The most common ones are:

Making of chatbots: Integrating LLMs into chatbots enables them to extract more precise answers directly from company documents and knowledge repositories. These chatbots are deployed for automating customer support services and following up on website leads, swiftly addressing inquiries and resolving issues.

Implementation of Advanced Search Engines: Integrating LLMs into search engines to enhance search results with answers generated by LLMs can improve the accuracy of informational queries, making it simpler for users to locate the information necessary for their tasks.

Query over company's data: Utilizing company data as context for LLMs enables employees to access answers to their queries effortlessly, covering areas such as HR inquiries about benefits, policies, as well as security and compliance matters.

B. What are the perks of RAG?

latest and accurate information: RAG ensures that an LLM's response is not solely reliant on outdated, fixed training data. Instead, the model leverages current external data sources to deliver responses.

Decreasing erroneous answers or hallucinations : RAG aims to reduce the chances of LLM models providing incorrect or fabricated information, often referred to as hallucinations, by basing their outputs on pertinent external knowledge. These outputs may also include references to original sources, facilitating human verification.

domain-specific responses: With RAG, the LLM can offer responses that are contextually appropriate and customized to an organization's unique or domain-specific data.

Operating in a manner that is both efficient and cost-effective : When it comes to customizing LLMs with domain-specific data, RAG stands out as a straightforward and cost-efficient approach. Organizations can implement RAG without the need for extensive model customization, which is particularly advantageous when frequent updates with new data are required.

IV. PROPOSED METHODOLOGY

Our approach involved training Google's Gen AI model with a crop recommendation methodology consisting of refining and optimizing the performance of Google's Gen AI model by incorporating crop recommendations and utilizing Google Maps data obtained through the Serp API. This allowed us to enhance the accuracy and effectiveness of the AI-powered system, enabling it to provide more precise and relevant crop-related information to users.

A. Getting nearby seed dealers

- a) We utilized The Serp API to query data from <https://www.google.com/maps>.
- b) This enabled us to retrieve geographical information and map-related data.:
- c) The data obtained directly from Google Maps enhanced the functionality and capabilities of our research or application.:
- d) Utilizing the Serp API facilitated the seamless integration of real-time map data into our system.:
- e) The integration of Google Maps data improved the accuracy and relevance of user information. :
- f) Overall, leveraging the Serp API and Google Maps data significantly contributed to the effectiveness and precision of our AI-powered system for crop recommendations. :

B. Getting crop and condition Recommendations

- a) We used a CSV with labels like nitrogen, phosphorus, potassium, humidity, and temperature.:
- b) By using text splitter, Google Embeddings, and Vector DB we fed it to GenAI.:
- c) Like this, we made a retrieval model that helped us get ideal conditions from crop and conditions to crop.:

C. Using Low-code approach

- a) Flowise is an open-source low-code tool for developers to build customized LLM orchestration flow & AI agents. :
- b) Document loaders allow you to load documents from different sources like PDF, TXT, CSV, Notion, Confluence, etc. They are often used together with :

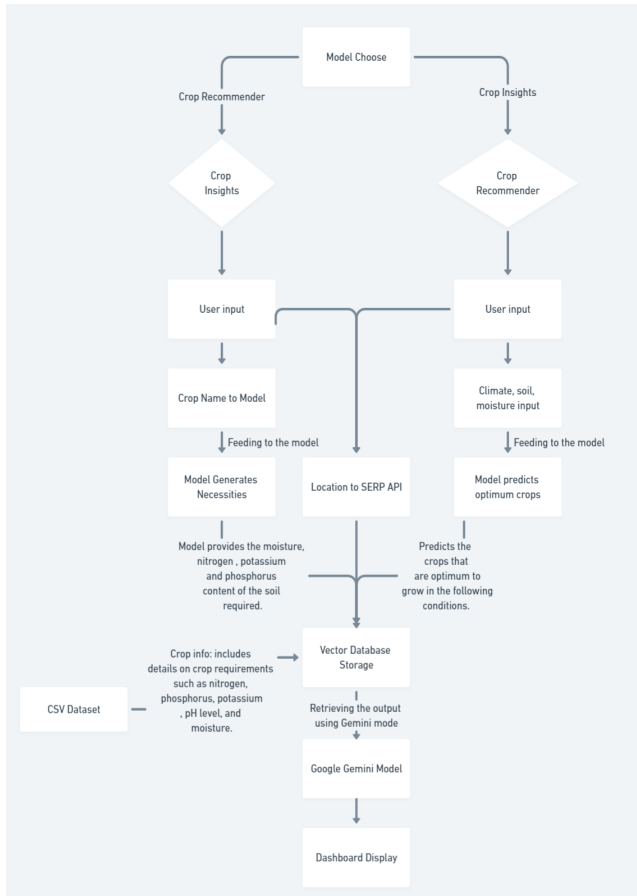


Fig. 2. Proposed Methodology

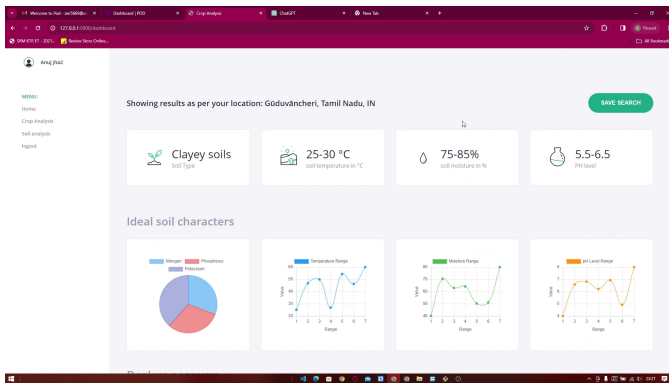


Fig. 3. Dash Board showing result for rice

c) Vector Stores are to be upserted as embeddings, which can then be retrieved upon query.:

d) Apart from this Flowise also allowed us to feed data scrapped by Serp-API into VectorDB and used in our retrieval model powered by Google's GenAI:

D. Using Jinja and Flask for Presenting the web application

We leveraged the power of Flowise to develop a sophisticated low-code model tailored to agricultural needs. Through a custom API integration, we seamlessly retrieved data outputs

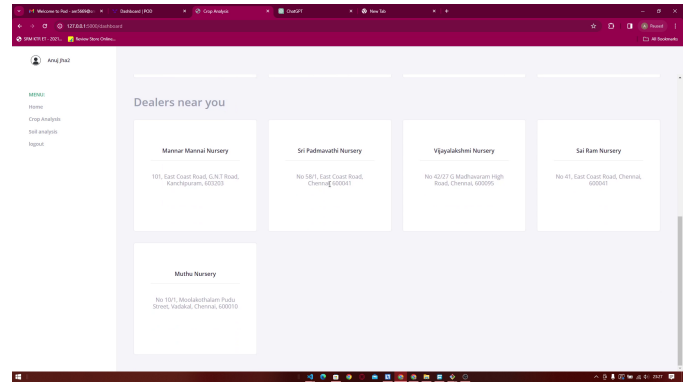


Fig. 4. showing near by location of the seed dealers

from Flowise, enabling us to incorporate its insights directly into our project's decision-making processes.

To manage user interactions and data securely, we opted for Supabase, a robust platform that not only handles user registration and authentication but also provides a scalable and reliable data storage solution.

Our user-centric approach allows individuals to select from a range of options tailored to their specific needs. They can either receive detailed guidance on ideal growing conditions for a chosen crop, maximizing their agricultural success, or explore recommendations for crops that align with their land characteristics, optimizing productivity and yield. Underpinning this functionality is a meticulously crafted API architecture built on Flask, ensuring smooth navigation and efficient data exchange. Complementing this backend framework is a visually engaging frontend developed using Jinja, delivering a seamless user experience that blends functionality with intuitive design.

V. FINE-TUNING METHODOLOGY FOR MODEL OPTIMIZATION

Unlocking Agriconnect's full potential relies on fine-tuning a powerful language model named Gemini. This section dives into how we've fine-tuned Gemini to work its magic in two key areas of our platform: Crop Insights and Crop Recommendations. We'll delve into the unique data sources and training methods we've used to customize Gemini's responses for each model. This tailored approach allows Gemini to provide farmers with precise insights tailored to their location and the crops they're interested in.

A. Crop Insights Model

The Crop Insights Model is the backbone of Agriconnect's data gathering and retrieval system. To provide farmers with spot-on insights tailored to their location, we've perfected a fine-tuning technique for Gemini, our powerful language model.

- First, we start with a robust CSV file packed with essential crop details. Optimal moisture levels, temperature ranges, and key soil nutrients like nitrogen, phosphorus,

and potassium. This data gives Gemini a solid understanding of what crops need to thrive.

- Next, we incorporate a fine-tuned prompt to guide Gemini's responses. This prompt ensures that the retrieved information is tailored to the user's specific location. To achieve this, Agriconnect leverages the power of SerpApi. This API seamlessly integrates with Google Maps, enabling the platform to extract location-specific data based on user searches. This data is then fed into a vector database, which acts as a centralized repository for location-specific information.
- Furthermore, the Crop Insights Model takes advantage of a pre-built CSV file meticulously created by the development team. This file, meticulously categorized by state, provides crucial details about seed and fertilizer dealers across various regions. This dealer information is also incorporated into the vector database, enriching Gemini's knowledge base with valuable resources relevant to the user's location.

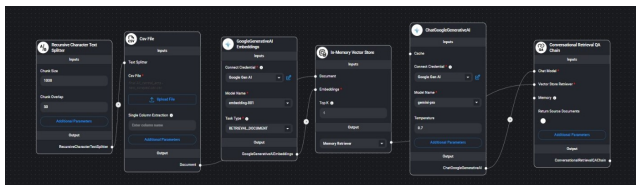


Fig. 5. CSV retrieval model

B. Crop Recommender Model

The Crop Recommender Model leverages the power of the fine-tuned Gemini LLM to deliver personalized crop recommendations to farmers. This section delves into the specific approach used to optimize Gemini for this critical functionality.

Unlike the Crop Insights Model, the focus here shifts towards user-provided data. Farmers actively participate by inputting crucial information about their location and specific environmental conditions. This data encompasses:

- **Climate:** Farmers can specify details like temperature ranges, humidity levels, and average rainfall patterns in their region.
- **Soil Type:** Our system now enables users to specify the dominant soil type in their fields, such as sandy loam, clay, or black soil. This information is vital for crop management decisions, as different soil types have varying water retention capacities and nutrient levels.
- **Soil Temperature:** Additionally, users can also input the current soil temperature. This data is essential for understanding optimal crop growth conditions, as soil temperature affects nutrient availability and root development. By considering soil temperature alongside other factors, farmers can make more informed decisions to enhance their crop yields.

- **Soil Moisture:** Farmers can provide data on the current moisture content of their soil, which is a crucial factor for optimal crop growth.

This user-generated data is then fed into the fine-tuning process for Gemini. However, it's important to note that Gemini isn't solely reliant on this new information. The fine-tuning process also leverages the pre-existing knowledge base established through the user-created CSV file used in the Crop Insights Model. This CSV, meticulously categorized by state, provides valuable information about crop requirements and their suitability for different soil types.

By combining user-specific data with the broader knowledge base from the CSV file, the fine-tuning process empowers Gemini to deliver highly personalized crop recommendations. Gemini can analyze the user's location, climate, soil type, and moisture content, and then recommend crops that are best suited for those specific conditions. This personalized approach goes beyond generic crop data recommendations, ultimately offering farmers a valuable tool for optimizing their crop selection and maximizing yield potential.

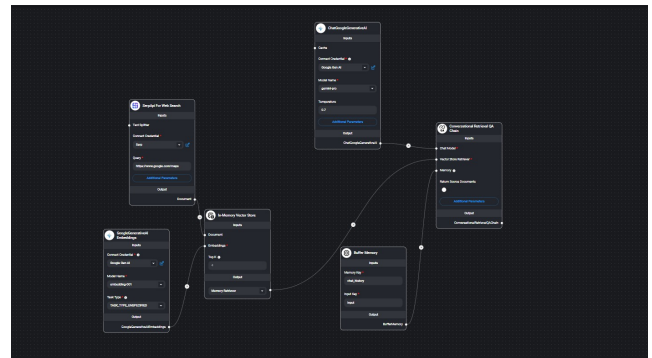


Fig. 6. Dealers Retrieval Model

C. Dataset Description

A crucial dataset for our project resides in a table format. This data provides valuable insights into the factors influencing rice crop production. Each data point details the amount of nitrogen, phosphorus, and potassium potentially applied, alongside environmental conditions like temperature (in degrees Celsius), humidity (as a percentage), rainfall (likely in millimeters), and soil acidity/alkalinity (pH). Importantly, each data point is labeled as "rice," indicating its relevance to our project. This dataset allows us to analyze the relationships between these environmental factors and rice growth. Researchers can leverage this information to understand how temperature, humidity, and rainfall influence rice yields. Additionally, the data can be used to create models that predict optimal nutrient levels (nitrogen, phosphorus, and potassium) for various environmental conditions, ultimately contributing to optimized rice crop production.

- **Columns:**
- **N(Nitrogen):** Numerical value, representing the amount of nitrogen applied.

- **P(Phosphorus):** Numerical value, representing the amount of phosphorus applied.
- **K(Potassium):** Numerical value, representing the amount of potassium applied.
- **temperature:** Numerical value, representing the temperature in degrees Celsius.
- **humidity:** Numerical value, representing the relative humidity as a percentage.
- **ph:** Numerical value, representing the acidity or alkalinity of the soil.
- **rainfall:** Numerical value, representing the amount of rainfall in millimeters.
- **label:** Categorical value, representing the crop type.

Overall, this dataset provides valuable insights for researchers, agricultural scientists, and data analysts interested in optimizing rice crop production and understanding the influence of environmental factors.

D. Vector Database

Throughout Agriconnect’s functionalities, a key component underpins the system’s effectiveness: the vector database. This specialized database plays a critical role in storing and managing the vast amount of information used to fine-tune Gemini, the large language model at the heart of the platform.

The vector database offers several advantages over traditional relational databases for our application.

- **Efficient Data Representation:** Unlike traditional row-and-column structures, the vector database stores information as high-dimensional vectors. This allows for efficient storage and retrieval of complex data, such as the relationships between crops, environmental factors, and dealer locations.
- **Fast Similarity Search:** A core strength of vector databases is their ability to perform rapid similarity searches. This is essential for Agriconnect, as it allows Gemini to quickly identify crops with requirements that closely match the user’s specific climate, soil characteristics, and location.
- **Scalability:** As Agriconnect’s user base grows, the vector database can seamlessly scale to accommodate the increasing volume of data. This ensures efficient performance and maintains the responsiveness of the platform.

In summary, the vector database acts as the central nervous system of Agriconnect’s knowledge base. It facilitates the intricate dance between user-provided data, the pre-defined knowledge base from the CSV file, and the fine-tuning process for Gemini. This efficient management of information ultimately empowers Gemini to deliver personalized crop recommendations and location-specific insights, aiding farmers in making informed decisions for optimized crop production.

VI. RESULT

Agriconnect’s user interface features a dynamic dashboard that gives farmers practical insights. This section explains how the dashboard combines the Crop Insights Model and

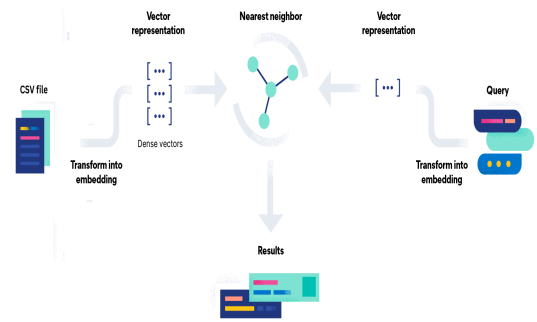


Fig. 7. Working of the vector database

the Crop Recommender Model to provide easy-to-understand information.

A. Visualizing Crop Requirements:

- **Crop Recommendations:** The dashboard prominently displays the crops best suited for the user’s specific location, climate, and soil characteristics. This information is generated by the Crop Recommender Model, which analyzes user-provided data and leverages its fine-tuned knowledge base.
- **Visualizations and Data Representation:** To enhance comprehension, the dashboard utilizes various charts, graphs, or other visual elements. These may depict optimal temperature ranges, required moisture levels, or nutrient requirements for the recommended crops. This allows farmers to quickly grasp the critical factors for successful crop growth.

B. Facilitating Resource Acquisition:

- **Dealer Information Integration:** The dashboard seamlessly integrates the functionalities of the Crop Insights Model. By leveraging the user’s location and the dealer information stored in the vector database, the dashboard displays a list of nearby seed and fertilizer dealers.
- **Contact Details:** This list goes beyond simply providing dealer names. It incorporates crucial contact information such as addresses and phone numbers. This empowers farmers to easily locate and contact relevant resources, eliminating the need for separate searches and facilitating a streamlined process.

Agriconnect’s dashboard acts as a one-stop shop for farmers. It not only provides personalized crop recommendations based on their specific conditions but also removes the barrier of resource acquisition by providing readily accessible dealer information. This combination empowers farmers to make informed decisions and take action based on the insights gleaned from the platform.

VII. CONCLUSION

The agricultural landscape is changing fast. With technology advancing and the environment shifting, farmers face new

challenges. That's where Agriconnect steps in. It's more than just a tool; it's a game-changer, revolutionizing how farming is done.

We've seen firsthand how Agriconnect is transforming farmers' lives. It's not just about boosting yields and cutting costs; it's about empowering farmers with the knowledge and support they need to thrive in a complex world.

But Agriconnect goes beyond numbers. It's about fostering a deeper bond between farmers and their land, encouraging sustainable practices that ensure the future of farming.

Looking ahead, Agriconnect's potential knows no bounds. By staying innovative and responsive to farmers' needs, it can make a global impact, guiding agriculture toward a brighter, more sustainable future.

Let's embrace Agriconnect as a symbol of hope and possibility. Together, we can create a future where farming isn't just about survival, but about abundance and joy for generations to come.

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